

Web Programming in Haskell

In this final article in the series on Haskell, the author explores how to use it for Web programming.



Scotty is a Web framework written in Haskell, which is similar to Ruby's Sinatra. You can install it on Ubuntu using the following commands:

```
$ sudo apt-get install cabal-install
$ cabal update
$ cabal install scotty
```

Let us write a simple 'Hello, World!' program using the Scotty framework:

```
-- hello-world.hs

{-# LANGUAGE OverloadedStrings #-}

import Web.Scotty
main :: IO ()
main = scotty 3000 $ do
  get "/" $ do
    html "Hello, World!"
```

You can compile and start the server from the terminal using the following command:

```
$ runghc hello-world.hs
Setting phasers to stun... (port 3000) (ctrl-c to quit)
```

The service will run on port 3000, and you can open `localhost:3000` in a browser to see the 'Hello, World!' text. You can then stop the service by pressing *Control-C* in the terminal. You can also use Curl to make a query to the server. Install and test it on Ubuntu as shown below:

```
$ sudo apt-get install curl
$ curl localhost:3000
Hello, World!
```

You can identify the user client that made the HTTP request to the server by returning the 'User-Agent' header value as illustrated in the following example:

```
-- request-header.hs
```